Strain

National 5 · Mechanisms & structures · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 A 10 m wire stretches 5.0 mm when a force is applied. Calculate the strain.

- ε = 5 ÷ 10 000 = 0.0005

2 The allowable strain on a bar is 0.0075 and its length is 2.5 m. Calculate the change in length.

- ΔL = ε × L = 0.0075 × 2500 = 18.75 mm (19 mm)

3 During testing a steel rod stretches 0.6 mm and the strain is 0.0012. Calculate the original length of the rod.

- L = ΔL ÷ ε = 0.6 ÷ 0.0012 = 500 mm